Sekh Mainul Islam

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Education

- **University of Copenhagen** — Copenhagen, Denmark
  PhD Fellow, Department of Computer Science — *September 2024 – August 2027 (expected)*
  - Supervisors: Prof. Isabelle Augenstein & Prof. Pepa Atanasova
  - Research Areas: Interpretability, Natural Language Processing
- **Indian Institute of Technology Kharagpur (IIT KGP)** — Kharagpur, India
  M.S. by Research, Computer Science & Engineering; CGPA: 9.49/10 — *January 2019 – September 2022*
  - Supervisor: Prof. Sourangshu Bhattacharya • Research Areas: Interpretability, NLP
  - *Thesis:* "A Knowledge-Driven Approach to Aspect-Level Sentiment Classification"
- **Maulana Abul Kalam Azad University of Technology** — Kolkata, India
  B.Tech., Computer Science & Engineering; CGPA: 9.13/10 — *July 2014 – May 2018*

Publications

- **Explaining the Roles of Experts in Mixture-of-Experts LLMs for Personalization**: **Sekh Mainul Islam**, Pepa Atanasova, Isabelle Augenstein. *Preprint.* [Paper]
- **Shaping the Document, Not the Model: Optimized Suffixes for Context Following**: Gia-Bao Dinh Ho, **Sekh Mainul Islam**, Masahiro Kaneko, Timothy Baldwin, Isabelle Augenstein. *Preprint.* [Paper]
- **Multi-Step Knowledge Interaction Analysis via Rank-2 Subspace Disentanglement**: **Sekh Mainul Islam**, Pepa Atanasova, Isabelle Augenstein. *EMNLP 2026 Findings.* [Paper]
- **BiasGym: A Simple and Generalizable Framework for Analyzing and Removing Biases through Injection**: **Sekh Mainul Islam**, Nadav Borenstein, Siddhesh Milind Pawar, Haeun Yu, Arnav Arora, Isabelle Augenstein. *EMNLP 2026 Findings.* [Paper]
- **Evaluation Framework for Highlight Explanations of Context Utilisation in Language Models**: Jingyi Sun, Pepa Atanasova, Sagnik Ray Choudhury, **Sekh Mainul Islam**, Isabelle Augenstein. *Computational Linguistics*, 2026. [Paper]
- **AR-BERT: Aspect-Relation Enhanced Aspect-Level Sentiment Classification with Multi-Modal Explanations**: **Sk Mainul Islam** and Sourangshu Bhattacharya. *WWW 2022.* [Paper]
- **Fair Data Generation using Language Models with Hard Constraints**: **Sk Mainul Islam**, Abhinav Nagpal, Balaji Ganesan, Pranay Kumar Lohia. *CtrlGen Workshop at NeurIPS 2021.* [Paper]

Research & Professional Experience

- **New York University Abu Dhabi** — Abu Dhabi, UAE
  Research Engineer (Supervisor: Prof. Nasir Memon) — *November 2022 – August 2024*
  - Investigated the effect of concept drift on misinformation detection classifiers over time.
  - Identified two key distributional shift factors: rapidly evolving social media topics and shifting linguistic characteristics (vocabulary, writing style), which degrade classifier performance.
  - Proposed a novel adaptation framework combining efficient human-in-the-loop labelling with continual learning to re-calibrate classifiers on new distributional data.[Paper]
- **Siemens Technology and Services Pvt. Ltd.** — Bangalore, India
  Research Engineer — *July 2022 – October 2022*
  - Developed a proof-of-concept pipeline for automated structured data extraction and knowledge representation from OCR-processed industrial device manuals.
  - Addressed challenges in cross-device generalisation and structural information extraction from engineering drawings.
- **Flipkart** — Bangalore, India
  Research Intern — *July 2021 – September 2021*

- Designed a clustering-based pipeline for relation surface-form extraction between e-commerce concepts from product review text.
- Fine-tuned a BERT-based self-supervised model on internal lifestyle-domain data and mapped extracted surface forms to Wikidata relations.[Paper]

- **IBM Research India** — Bangalore, India
  Research Intern — *November 2020 – June 2021*
  - Built a fairness-constrained language model for generating personal data entities in unstructured text.
  - Validated model effectiveness through behavioural, fairness, and adversarial testing; work accepted at *CtrlGen Workshop, NeurIPS 2021.*[Paper]

- **FOSSEE, IIT Bombay** — Mumbai, India
  Software Developer Intern — *Summer 2017*
  - Developed a browser-based port of Xcos (a graphical simulator), eliminating the need for local software installation.

Research Projects

- **Dynamic High-Level E-Commerce Concept Mining** — IIT KGP & Flipkart
  Academic Collaboration — *February 2021 – June 2022*
  - Identified limitations of existing concept extraction methods in capturing long-tail and composite e-commerce concepts.
  - Introduced a constrained soft-clustering model to learn composite and non-trivial concepts from product data.

- **Debiasing Language Models via Prompt Design** — Stanford University
  Fatima Fellowship — Mentors: Abubakar Abid & Firas Abuzaid — *April 2021 – December 2021*
  - Estimated attribute-specific bias subspaces using controlled attribute-word swapping and contrastive PCA.
  - Proposed a post-hoc debiasing method that subtracts the estimated bias subspace from sentence representations while preserving desirable semantic content.

Professional Activities

- **Reviewer:** NeurIPS 2022; EMNLP 2021, 2020; NAACL 2021; EACL 2021; *SEM 2021; CODS-COMAD 2024.

Teaching Experience

- **Teaching Assistant, IIT Kharagpur:** Foundations of Algorithm Design & Machine Learning (Spring 2022, 2021); Computing Lab (Autumn 2021); Scalable Data Mining (Autumn 2020).
- **Teaching Assistant, Centre for AI, IIT Kharagpur:** Professional Course in Machine Learning (2019).

Talks, Schools & Honours

- Attended LxMLS 2025 Lisbon Machine Learning School (Lisbon, Portugal).
- Attended ALPS 2025 Winter School on Speech and NLP (Grenoble, France).
- Attended HPLT & NLPL 2025 Winter School on Pretraining Data Quality and Multilingual LLM Evaluation (Oslo, Norway).
- Selected (with full funding) for the Cornell–Maryland–Max Planck Pre-doctoral Research School 2022.
- Invited attendee, Amazon Research Day 2023 and 2019 (Amazon India, Bangalore).

Technical Skills

Languages:	Python, C/C++, Java, Spark/Scala, MATLAB
Frameworks:	PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn, Pandas, NumPy, NLTK

Selected Coursework

Deep Learning • Machine Learning • Scalable Data Mining • Information Retrieval • Algorithm Design